Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 06)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
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9	
10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 Explain how the control of emission **location** and emission **timing** can reduce the severity of pollution. Use examples.

[6 marks]

Emission location .....

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Emission timing .....

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0 1 . 2 Explain the **polluter pays** principle and suggest why it encourages pollution prevention.

[2 marks]

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0 1 . 3 Explain the **precautionary principle** as applied to the release of a new chemical.

[2 marks]

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0 2 Table 1 shows some radioactive isotopes, their half-lives and the type of radiation they emit.

Table 1

Isotope	Half-life	Main radiation	Source
Iodine-131	8 days	Beta / gamma	Fission in reactors
Strontium-90	29 years	Beta	Fission in reactors
Caesium-137	30 years	Beta / gamma	Fission in reactors
Plutonium-239	24 400 years	Alpha	Neutron bombardment of U-238
Uranium-238	4.5×10^9 years	Alpha	Occurs naturally in rocks

0 2 . 1 Define the term **half-life**.

[1 mark]

0 2 . 2 A sample of iodine-131 has an activity of 6400 Bq. Calculate its activity after 32 days.

[2 marks]

Answer _____ Bq

0 2 . 3 Use Table 1 to explain why isotopes with half-lives of several decades, such as caesium-137, cause the greatest concern after a nuclear accident.

[3 marks]

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0 2 . 4 Explain why plutonium-239 is relatively safe outside the body but very dangerous if inhaled or ingested.

[3 marks]

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0 2 . 5 Explain what is meant by an **activation product**.

[1 mark]

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0 3 Table 1 shows the mean annual soil loss measured on four fields of one farm. Soil forms on this farm at about 1 t ha^{-1} yr^{-1}.

Table 1

Field	Slope	Land use	Mean annual soil loss / t ha^{-1} yr^{-1}
A	2%	Permanent grassland	0.8
B	9%	Maize, ploughed up and down the slope	24.0
C	9%	Maize, contour ploughed	9.0
D	1%	Spring barley, soil left bare over winter	6.5

0 3 . 1 Use Table 1 to identify the fields that are suffering **accelerated** soil erosion, and explain your answer.

[2 marks]

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0 3 . 2 Calculate the percentage reduction in soil loss achieved by changing field B to contour ploughing.

[2 marks]

Answer ..... %

0 3 . 3 Explain how each of the following reduces erosion by water.

[3 marks]

Contour ploughing .....

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Terracing .....

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Rows of stones .....

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0 3 . 4 Field D is exposed to strong winds. Explain **four** methods, other than planting hedgerows, that would reduce wind erosion there.

[4 marks]

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0 3 . 5 Describe **four** impacts of the eroded soil **away from** the fields where it is lost.

[4 marks]

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0 4 A student sampled the ground-flora of two woodlands. Table 1 shows the results for woodland A.

Table 1

Species	Number of individuals (n)
Bluebell	40
Wood anemone	30
Dog's mercury	20
Bramble	10
Total (N)	100

0 4 . 1 Use the formula $D = N(N - 1) / \Sigma n(n - 1)$ to calculate Simpson's index of diversity for woodland A.

[3 marks]

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Answer

0 4 . 2 Woodland B had an index of 1.6. State what this shows and give **one** limitation of using the index alone to compare the woodlands.

[2 marks]

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0 5 Figure 1 shows a simplified nitrogen cycle with some processes labelled 1 to 6.

Figure 1

Number	Process description
1	Nitrogen gas converted to ammonium ions by Rhizobium in root nodules
2	Ammonium ions oxidised to nitrite then nitrate by Nitrosomonas and Nitrobacter
3	Nitrates reduced to nitrogen gas by Pseudomonas in anaerobic soil
4	Amino groups in dead organic matter released as ammonium ions by decomposers
5	Nitrogen and oxygen react during lightning to form NOx
6	Nitrates washed out of soil into rivers

0 5 . 1 Name processes 1 to 6.

[3 marks]

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0 5 . 2 Explain why draining a waterlogged field increases the nitrate available to crops.

[3 marks]

0 5 . 3 Explain why nitrogen is described as a resource that is 'not very abundant' for living organisms even though the atmosphere is 78% nitrogen.

[2 marks]

0 5 . 4 Explain why growing legumes such as clover in a crop rotation reduces the need for artificial fertiliser.

[2 marks]

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0 6 . 1 Describe the process of anaerobic sludge digestion and give **two** benefits of the process.

[4 marks]

0 6 . 2 Explain **two** problems of using sewage sludge as an agricultural fertiliser and how each can be reduced.

[4 marks]

0 6 . 3 Explain why it is more efficient to have separate collection systems for foul sewage and for surface drainage water.

[2 marks]

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0 7 . 1 Which of the following water sources is usually the most convenient for large communities? Tick one box.

[1 mark]

A Desalinated seawater ☐

B River water ☐

C Rainwater catchment ☐

D Newly discovered aquifers ☐

0 7 . 2 Explain why the quality requirements for water differ between uses, using **two** named uses.

[2 marks]

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0 7 . 3 Explain why inter-basin transfers may be used and give **one** disadvantage.

[2 marks]

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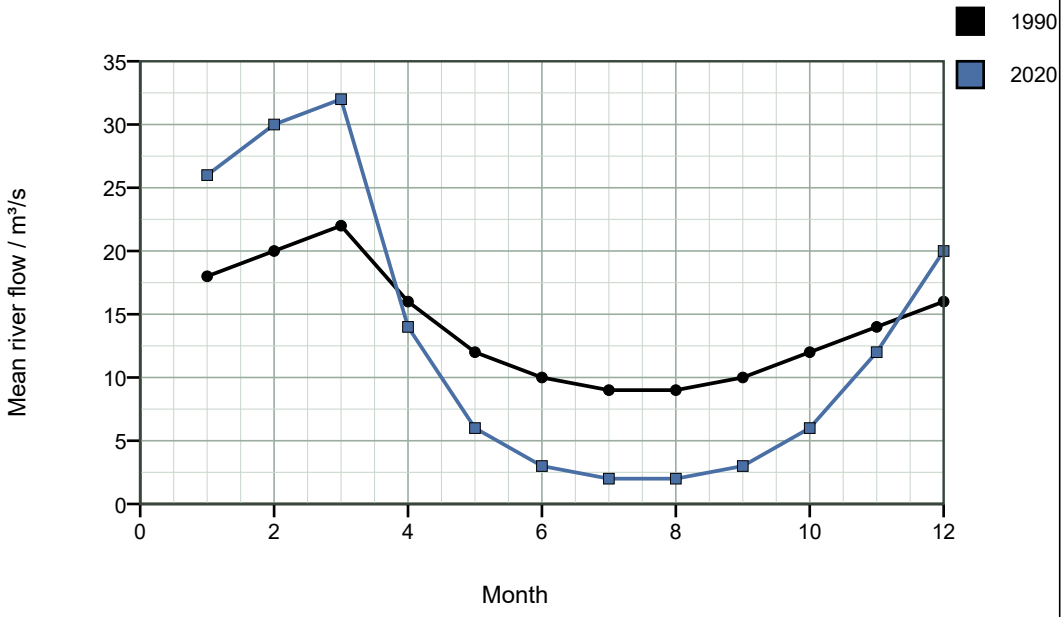
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0 8 Figure 1 shows the monthly flow of a river in South America that is fed partly by glacier meltwater, in 1990 and 2020. The glacier feeding the river shrank considerably between the two dates.

Figure 1



0 8 . 1 Compare the river flow in 1990 and 2020.

[3 marks]

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0 8 . 2 Explain how the shrinking of the glacier explains the differences.

[3 marks]

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0 8 . 3 Suggest **two** problems the change in flow causes for people living downstream.

[2 marks]

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0 8 . 4 Explain why a glacier can be extending down its valley yet still be losing ice volume.

[2 marks]

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0 9 Table 1 shows the bypass ratio and first year of use of jet engines used on four airliners.

Table 1

Engine	Aircraft	Bypass ratio	First use
RR Olympus	Concorde	0 : 1	Early 1950s
RR Conway	Boeing 707	0.25 : 1	Late 1950s
P&W JT9D	Boeing 747	5 : 1	1969
RR Trent 1000	Boeing 787	10 : 1	2011

0 9 . 1 Explain how a high bypass ratio reduces engine noise.

[3 marks]

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0 9 . 2 Describe **two** other features of aircraft design that reduce noise.

[2 marks]

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2 .....

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0 9 . 3 Explain how each of the following airport operating procedures reduces noise nuisance.

[4 marks]

Constant descent angle .....

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Night flight restrictions .....

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Noise-based landing charges / quota count .....

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Multiple runways .....

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0 9 . 4 Explain why aircraft are noisier on take-off than on landing.

[1 mark]

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1 0 . 1 Discuss how the features of energy resources determine their suitability for different uses and their environmental sustainability.

[9 marks]

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Extra space

1 0 . 2 State **one** reason why nuclear power stations can be located where transport infrastructure is poor.

[1 mark]

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Write an essay on one of the following topics.

11 . 1 Discuss how the atmosphere supports life on Earth and evaluate the extent to which human activities threaten these life-support functions.

[25 marks]

OR

11 . 2 Discuss the strategies that can be used to provide sustainable supplies of water for a growing population.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS